Yuchen Zhang

yuchen.zhang@uth.tmc.edu | Website | Google Scholar

Research Interests

Continual Learning, Reinforcement Learning, Causal Inference,
Multimodal Representation Learning

Education

The University of Texas Health Science Center at Houston, *Ph.D. Candidate in Biostatistics, Houston, TX* — Aug 2024 – Dec 2027 (Expected)

- **Advisor:** Lulu Shang
- **Thesis Proposal:** Machine Learning for Meaningful Discovery in Biomedical Data

Columbia University Mailman School of Public Health, *M.S. in Biostatistics, New York, NY* — Aug 2022 – May 2024

The Ohio State University, *B.S. in Applied Mathematics, Columbus, OH* — Aug 2019 – May 2022

Experience

Graduate Research Assistant, The University of Texas MD Anderson Cancer Center, Houston, TX — Apr 2026 – Present

- Developing multimodal representation learning methods using contrastive learning, transformer fusion, and spatial modeling for biomedical data integration.
- Integrating large language models into an internal data foundry for biological discovery.

Graduate Research Assistant, The University of Texas Health Science Center at Houston, Houston, TX — Oct 2024 – Apr 2026

- Applied causal inference models to uncover causal links between risk factors and Alzheimer's disease using large population datasets.

Publications

Filter Before Judging: Decision-Scope Evidence Projection for Robust LLM Judging
AIWILD@NeurIPS 2026
Yuchen Zhang, Gang Zhang, Yixiang Hong

Historical-Feature Residual Adaptation for Sequential Screening
AgenticLS@NeurIPS 2026
Yixiang Hong, **Yuchen Zhang**

MELON: Multimodal Learning Framework for Spatial Multimodal Omics Data Integration
AI for Science@ICML 2026
Yuchen Zhang, Chunru Lin, Liuqing Yang, Yuying Xie, Yu Shen, Lulu Shang

A path-specific effect approach to mediation analysis with time-varying mediators and time-to-event outcomes accounting for competing risks
Statistics in Medicine, 2026, 45(3–5):e70425
Arce Domingo-Relloso, **Yuchen Zhang**, Ziqing Wang, Astrid M. Suchy-Dicey, Dedra S. Buchwald, Ana Navas-Acien, Joel Schwartz, Kiros Berhane, Brent A. Coull, Linda Valeri

Uncovering the features of industrial odors-derived environmental complaints and proactive countermeasures by using machine-learning
Journal of Environmental Management, 2024, 370:122900
Hao Xiao, Jinping Tian, Yalin Chen, Chengwen Wang, **Yuchen Zhang**, Lyujun Chen

Presentations

2026 ADSP Annual Program Meeting 2026
National Institutes of Health, Rockville, MD

- Presented "Unsupervised deep learning-derived imaging phenotypes link brain structure, proteomics, and Alzheimer's disease risk."

STATGEN 2026 Conference on Statistics in Genomics and Genetics 2026
Atlanta, GA

- Delivered a talk in the Data Integration program.

2024 Biostatistics Practicum Symposium 2024
Columbia University Mailman School of Public Health

- Presented a simulation study in causal inference.

Math Cycle Conference 2022
The Ohio State University

- Presented mathematical modeling for the analemma (talk and poster).

Awards

Tinker Research Funds 2026
Thinking Machines Lab

NSF Travel Award 2026
Emory University

Robert H. Bigelow Endowed Scholarship 2026
UTHealth at Houston

Aurelia Murphy Killian Endowment in Public Health Award 2025
UTHealth at Houston

Outstanding Data Science Project 2023
Columbia University Mailman School of Public Health

- Project: World Life Expectancy by statistical classification.

Professional Service

Reviewer: ICASSP 2027; ICLR 2027; ENAR 2027; NeurIPS 2026; PLOS ONE; BMC; APHA 2026.

Teaching Assistant: Environmental Health GIS, Columbia University Mailman School of Public Health (2023).

Computing Experience

Languages/Systems: Python, C++, Java, SAS, SQL, R.

Statistics: Survival Analysis, Causal Inference, Longitudinal Analysis, Linear Regression.